INCH-POUND

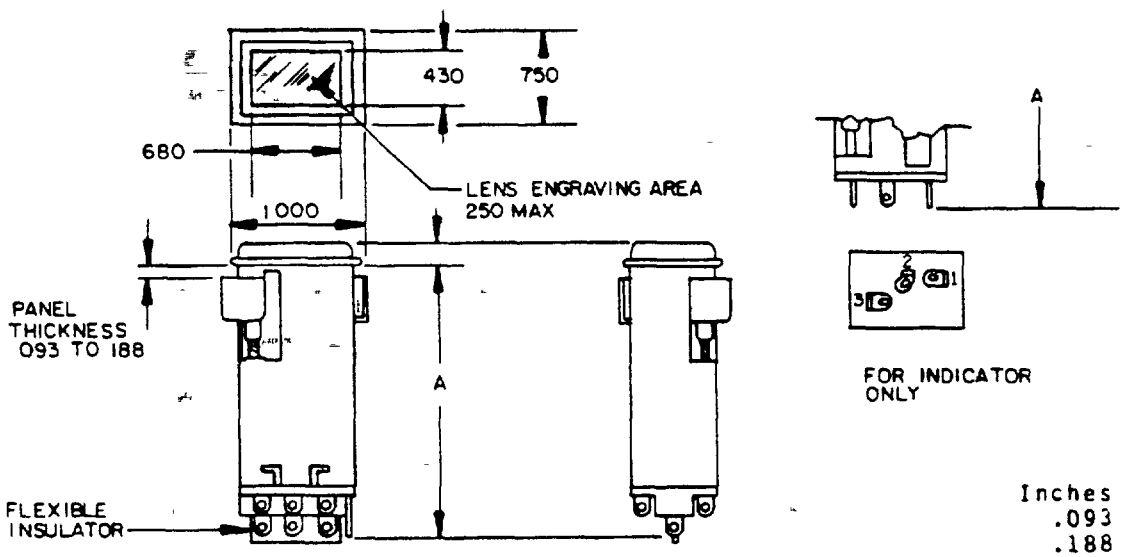
MIL-S-22885/58F
20 DECEMBER 1989
SUPERSEDING
MIL-S-22885/58E
10 September 1985

MILITARY SPECIFICATION SHEET

SWITCH, PUSH BUTTON, ILLUMINATED, 2-LAMP, SOLID MOUNT, 3 AMPERES, UNSEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-S-22885.



Dimension A	
Momentary and alternate action switch	2.200 max (55.88)
Indicator only	1.12 max (28.4)

Inches	mm
.093	2.36
.188	4.78
.250	6.35
.430	10.92
.655	16.64
.680	17.27
.750	19.05
.905	22.99
1.000	25.40
1.12	28.4
2.200	55.88

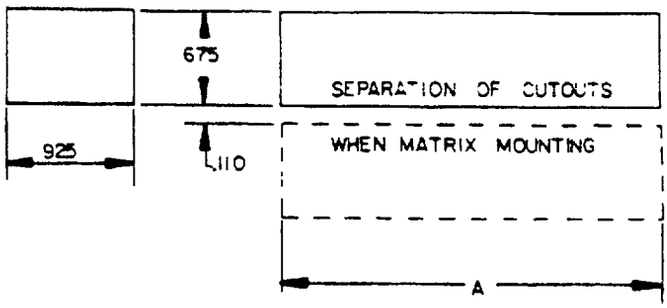
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. On 2 PDT switches, terminals are furnished in 2 center positions only.
5. The design configuration is optional within envelope dimensions shown.
6. Terminals shall be permanently identified as shown.

(F) denotes change

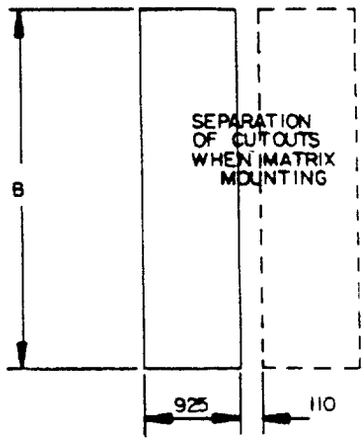
FIGURE 1 Switch, 2-lamp, solid mount.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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A HORIZONTAL MOUNTING

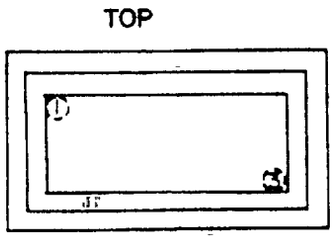


B VERTICAL MOUNTING

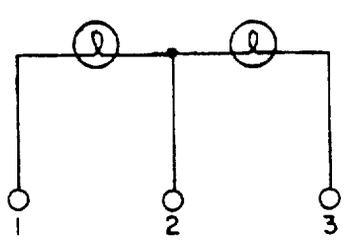
No. Units	Dimensions	
	A	B
1	.925 (23.50)	.675 (17.14)
2	1.925 (48.90)	1.425 (36.20)
3	2.930 (74.42)	2.180 (55.37)
4	3.935 (99.95)	2.935 (74.55)
5	4.940 (125.48)	3.690 (93.73)
6	5.945 (151.00)	4.445 (112.90)
7	6.950 (176.53)	5.200 (132.08)
8	7.955 (202.06)	5.955 (151.26)

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, the tolerance for three place decimals is ±.010 (0.25 mm)

FIGURE 2. Recommended panel outputs.



FRONT VIEW
NUMBER CORRESPOND TO
LAMP TERMINALS



LAMP SCHEMATIC

FIGURE 3. Lamp position and schematic.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 3 and tables I through VI.

Design and construction

Enclosure design. 1 (unsealed).

Switch or indicator unit shall consist of:

1 lamp module.

1 switch or indicator housing.

2 lamps, T-1-3/4, midget flanged base not included - order separately.

Lamp module: Includes lens assembly in accordance with MIL-S-22885/59 as specified (see table VI), holder, and divider when applicable (see table III). The lamps and lens assembly shall be capable of being installed or removed from the panel front without the use of special tools.

Positive grounding: Applicable when specified (see table VI).

Color and luminance values. In accordance with MIL-S-22885/12 (brightness is reduced by approximately 50 percent when EMI screen is installed).

Mounting flange: Front face of the mounting flange shall be corrosion-resistant steel with a natural lusterless finish.

Ratings. (Switch unit only.)

Electrical: See table II.

Operating force: 4.0 pounds maximum.

Weight: 2.0 ounces maximum.

Color. Shall appear white in the unlighted condition. In the lighted condition it shall appear in a color determined by the color filters (see MIL-S-22885/59).

Contact resistance: Not applicable after inductive load endurance tests.

Thermal shock: Temperature extremes shall be -55°C and +71°C (+55°C with lamps energized).

Vibration grade: 2 (10-500 Hz).

Part or Identifying Number (PIN): The PIN's shall be M22885/58- (dash number from table VI).

Qualification inspection:

Group submission The following applies when the momentary and alternate action switches are submitted for qualification at the same time.

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TABLE I. Qualification inspection (group submission).

Sample part number	Test	Extent of approval
M22885/58-31W	Qualification inspection table of MIL-S-22885	All part numbers
2 each· M22885/58-04W M22885/58-25W	Visual and mechanical inspection	

1/ Separate additional lamp modules shall be utilized for each color for group VIII tests.

TABLE II. Electrical ratings.

Load	Sea level	
	28 V dc (amperes)	115 V ac (amperes)
Resistive - - -	3.0	3.0
Inductive - - -	1.5	1.5
Lamp- - - - - -	1.5	1.0

TABLE III. Lens style.

Code	Lens style	
A	Solid	
B	Split horizontally	1 2
C	Split vertically	1 2

Ⓕ TABLE IV. Color filter symbols.

Color	Symbol
Blue	B
Green	G
Red	R
White	W
Yellow	Y

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Ⓕ TABLE V. Chromaticity limits and photometric brightness.

	Chromaticity limits 1/ 2360° Kelvin		Switch photometric brightness (footlamberts) 2/
Colors	x	y	
Green	.260	.570	25 footlamberts minimum
	.300	.630	
	.160	.660	
	.200	.720	
Yellow	.562	.415	300 footlamberts minimum
	.570	SL 3/	
	.596	.382	
	.605	SL 3/	
Red	.695	.285	80 footlamberts minimum
	.703	SL 3/	
	.655	.325	
	.660	SL 3/	
Blue	.150	.260	10 footlamberts minimum
	.220	.260	
	.220	.320	
	.150	.320	
White	.400	.420	150 footlamberts minimum
	.460	.420	
	.460	.380	
	.400	.380	

1/ The chromaticities of the diffusing lenses or colored filters expressed as x and y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2360°K.

2/ Switch photometric brightness, when viewed from the front of the applicable light housing.

3/ SL: Spectrum locus (where intersected by other of coordinate pair).

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TABLE VI. PIN's. 1/

Without EMI screen		With EMI screen			
M22885/58-		M232885/58-		Circuit characteristics and operation	Lens style code (see table III)
With front lens	Without front lens	With front lens	Without front lens		
01(X)	13(X)	31(X)	46(X)	2 PDT momentary	A
02(XX)	14(XX)	32(XX)	47(XX)	2 PDT momentary	B
03(XX)	15(XX)	33(XX)	48(XX)	2 PDT momentary	C
04(X)	16(X)	34(X)	49(X)	4 PDT momentary	A
05(XX)	17(XX)	35(XX)	50(XX)	4 PDT momentary	B
06(XX)	18(XX)	36(XX)	51(XX)	4 PDT momentary	C
07(X)	19(X)	37(X)	52(X)	2 PDT alternate	A
08(XX)	20(XX)	38(XX)	52(XX)	2 PDT alternate	B
09(XX)	21(XX)	39(XX)	54(XX)	2 PDT alternate	C
10(X)	22(X)	40(X)	55(X)	4 PDT alternate	A
11(XX)	23(XX)	41(XX)	56(XX)	4 PDT alternate	B
12(XX)	24(XX)	42(XX)	57(XX)	4 PDT alternate	C
25(X)	28(X)	43(X)	58(X)	Indicator	A
26(XX)	29(XX)	44(XX)	59(XX)	Indicator	B
27(XX)	30(XX)	45(XX)	60(XX)	Indicator	C

1/ PIN's are shown for units without color filters. When color filters are to be included, add the color symbols of table IV in place of (X) or (XX) with location sequence as shown in table III.

EXAMPLE. M22885/58- 03 R G

- 03: 2 PDT momentary switch with vertical split lens assembly.
- R: Red color filter on left side.
- G: Green color filter on right side.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities.
Army - AV, MI
Navy - AS
Air Force - 17, 99
DLA - ES

User activities:
Navy - MC, OS, SH
Air Force - 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5930-1427-01)

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